



JAL1603

TIME SERIES ANALYSIS AND FORECASTING

III Year - Semester VI

TIME SERIES ANALYSIS AND FORECASTING

A field of study that focuses on studying / analyzing data points collected sequentially over time and predicting future values based on the underlying historical patterns.

Applications in various domains:



TIME SERIES ANALYSIS AND FORECASTING

- **Time series Data** - a sequence of data points collected, recorded, or observed at successive points in time, often at equally spaced intervals.
- **Time Series Analysis** - helps uncover the structure and behavior of time-dependent data.
- **Time Series Forecasting** - uses the analyzed knowledge to make informed predictions about the future.

ASPECT	TIME SERIES ANALYSIS	TIME SERIES FORECASTING
Goal	Understand and interpret historical data	Predict future values
Techniques	Trend analysis, seasonality detection	ARIMA, LSTM, Exponential Smoothing
Focus	Decomposition and pattern recognition	Extrapolation of trends and patterns

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SYLLABUS

UNIT I

INTRODUCTION TO TIME SERIES AND FORECASTING

Introduction to Time Series: Different types of data - Internal structures of time series - Models for time series analysis - Autocorrelation and Partial Autocorrelation - Introduction to forecasting: The nature and uses of forecasting - Examples of Time series The Forecasting Process - Data for forecasting - Resources for forecasting.



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TIME SERIES AND FORECASTING

- The data in a time series - in the form of a sequence of quantitative observations about a system or process and made at successive points in time
- The points in time are equally spaced.
- Examples of time series data → gross domestic product, sales volumes, stock prices, weather attributes when recorded over a time spread of several years, months, days, hours, and so on.
- Successful utilization of time series data would lead to monitoring the health of the system over time. Eg. the performance of a company is tracked.

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TIME SERIES AND FORECASTING

- Time series analysis aims to utilize such data
 - To understand and interpret the underlying forces that produce the observed state of a system or process over time
 - To forecast the future state of the system or process in terms of observable characteristics
- Different statistical methods to explore and model the internal structures of the time series data → trends, seasonal fluctuations, cyclical behavior, and irregular changes.
- Several mathematical techniques and programming tools → to effectively design computer programs that can explore, visualize, and model patterns in time series data.

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DIFFERENT TYPES OF DATA

- Business analysts and data scientists come across many different types of data in their analytics projects.
- 3 categories of data most commonly found:
 1. Cross-sectional data
 2. Time series data
 3. Panel data
- Understanding what **type of data** is needed to solve a problem and what type of data can be obtained from available **sources** is important for formulating the problem and choosing the right methodology for analysis.

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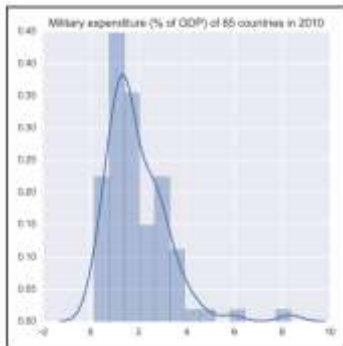
1. CROSS-SECTIONAL DATA

- Cross-sectional data or cross-section of a population is obtained by taking observations from multiple individuals at the same point in time.
- Comprise of observations taken at different points in time - in such cases time itself does not play any significant role in the analysis.
- Examples:
 - SAT scores of high school students in a particular year
 - Gross domestic product of countries in a year
 - Data for customer churn analysis
- Analysis starts with a plot of the variables
- They visualize central tendency, dispersion, skewness, and kurtosis.
- Examples:

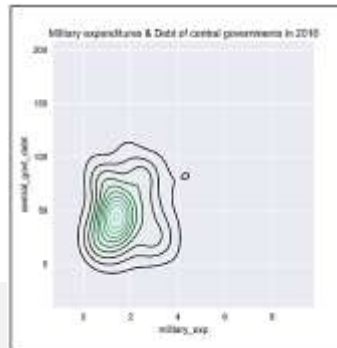
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1. CROSS-SECTIONAL DATA

- Eg. univariate analysis of military expenditure as a percentage of Gross Domestic Product of 85 countries in the year 2010.



- Eg. bivariate analysis considering total debt of the countries' central governments and their military expenditure in 2010.



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1. CROSS-SECTIONAL DATA

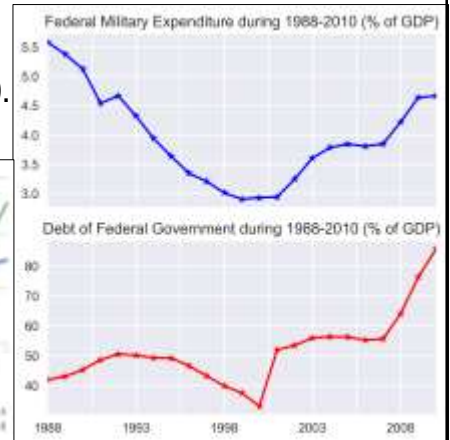
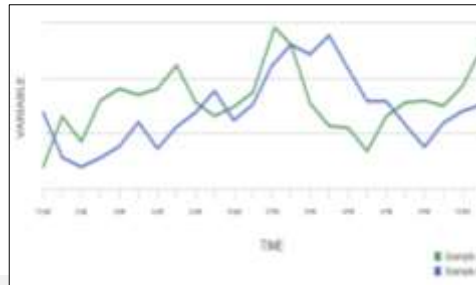
- Advanced methods → cross-sectional regression fit a linear regression model between several explanatory variables and a dependent variable.
- For example, in case of customer churn analysis, the objective could be to fit a logistic regression model between customer attributes and customer behavior described by churned or not-churned → explains the factors that make customers churn and can predict the outcome for a new customer.



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2. TIME SERIES DATA

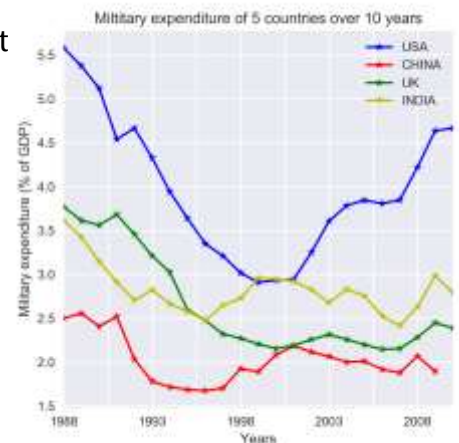
- Time series data is made up of quantitative observations on one or more measurable characteristics of an individual entity and taken at **multiple points in time**.
- Example: Military expenses and central government debt for India, for a span of 10 years from 2001 to 2010.
- Time series data characterized by several internal structures such as
 - ✓ trend
 - ✓ seasonality
 - ✓ stationarity
 - ✓ autocorrelation



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3. PANEL DATA

- Panel data → observing data from multiple entities over multiple points in time
→ also known as longitudinal data.
- Example: Military expenses and central government debt for four countries over the same period of 1960-2010.



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INTERNAL STRUCTURES OF TIME SERIES

- Special characteristics of time series data that requires its special mathematical treatment:
 - General trend
 - Seasonality
 - Cyclical movements
 - Unexpected variations
- Most time series has of one or more of the mentioned internal structures.
- Hence a time series can be expressed as a sum of the trend, seasonal, cyclical, and irregular components in that order.

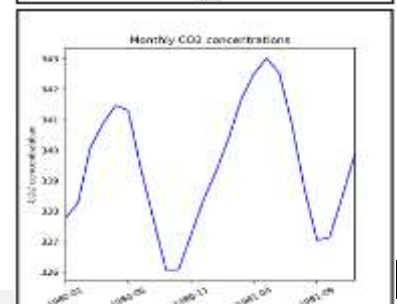
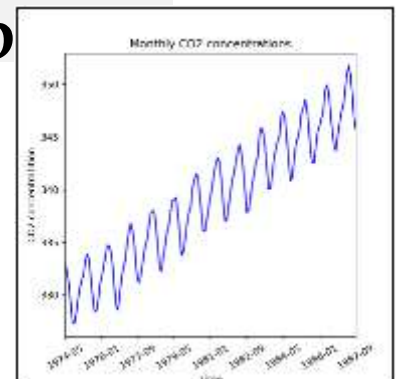
$$x_t = f_t + s_t + c_t + e_t$$

- $t \rightarrow$ the time index at which observations about the series have been taken at $t = 1, 2, 3 \dots N$ successive and equally spaced points in time.
- The objective of time series analysis $\rightarrow$ to decompose a time series into its constituent characteristics and develop mathematical models for each.
- Models $\rightarrow$ then used to **understand** what causes the observed behavior of the time series and to **predict** the series for future points in time.

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1. GENERAL TREND

- When a time series exhibits an upward or downward movement in the long run, it is said to have a general trend.
- Plot the data to check the presence of general trend in a data.
- Eg. CO₂ concentrations in air measured during 1974 through 1987
- Not be evident over a short run of the series.
- Requires data that dates significantly back in the past.
- The general trend in the time series is due to fundamental shifts or **systemic changes** of the process or system it represents.
- Modeled by setting up the time series as a **regression against time** and other known factors as explanatory variables.
- The **regression or trend line** $\rightarrow$ then used as a prediction of the long run movement of the time series.



2. SEASONALITY

- Seasonality demonstrates **repetitive and period variations** in a time series.
- Revealed by **EDA** (exploratory data analysis)
- Plot shows **periodic deviations**
- Eg. Peaks and troughs in the monthly sales volume of seasonal goods such as Christmas gifts or seasonal clothing.
- Practical techniques of determining seasonality
 - Run sequence plot
 - Seasonal sub series plot
 - Multiple box plots

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2. SEASONALITY

Run sequence plot

- A simple plot with time on x-axis and the variable on y-axis.
- Shows **movements in mean of the series, shifts in variance, presence of outliers**

Seasonal sub series plot

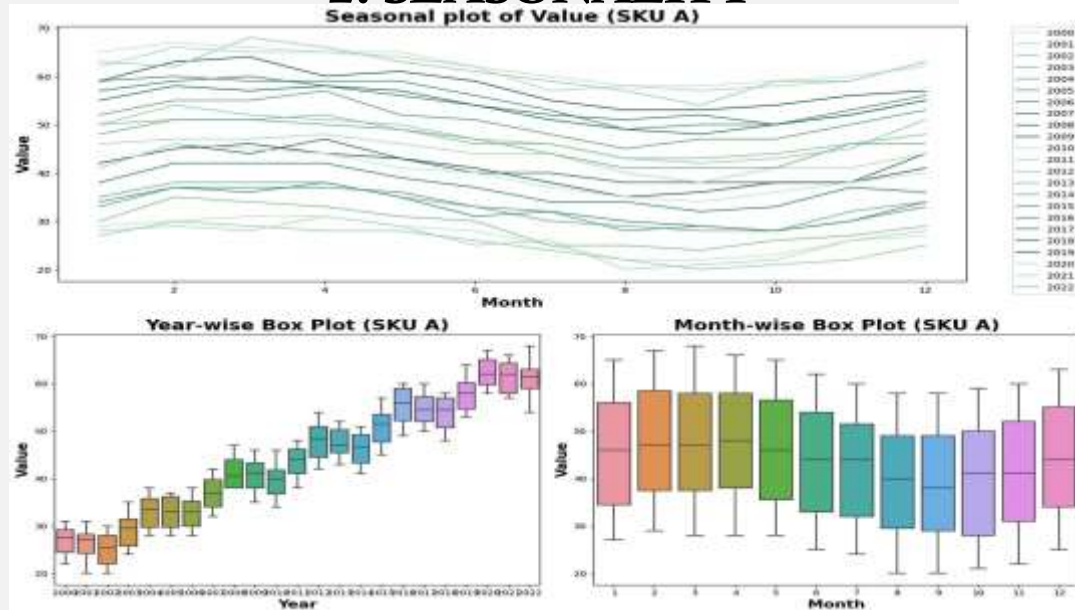
- Redraws the original series over batches of successive time periods, for a known periodicity of seasonal variations
- Reveals 2 properties:
 - Variations within seasons (as within a batch of successive months)
 - Variations between seasons (as between batches of successive months)

Multiple box plots

- Seasonal sub series plot made more informative when redrawn with seasonal box plots.
- Displays both central tendency and dispersion within the seasonal data over a batch of time units
- Separation between two adjacent box plots reveal the within season variations.

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2. SEASONALITY



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3. CYCLICAL CHANGES

- **Cyclical changes** → movements observed after every few units of time, but they occur less frequently than seasonal fluctuations.
- No fixed period of variations.
- The average periodicity for cyclical changes would be larger - most commonly in years (seasonal variations are observed within the same year).
- A long run plot of the time series is required to identify cyclical changes that can occur - as repetitive crests and troughs.
- Eg. time series related to economics and business
- Require data that dates significantly back in the past.
- Inflation of consumer price index (CPI) for India and USA over the period of 1960 through 2016.



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4. UNEXPECTED VARIATIONS

$$x_t = f_t + s_t + c_t + e_t$$

- In spite of being able to account for the three other components, there still exists an irreducible error component e_t
- It is random
- It does not exhibit systematic dependency on the time index.
- Unexpected variations are **stochastic** and cannot be framed in a mathematical model for a definitive future prediction
- This error is due to **lack of information** about explanatory variables that can model these variations or due to presence of a **random noise**.

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MODELS FOR TIME SERIES ANALYSIS

- Purpose of time series analysis → to develop a mathematical model that can explain the observed behavior of a time series and forecast the future state of the series.
- General models often used as building blocks of time series analysis:
 - Zero mean models
 - Random walk
 - Trend models
 - Seasonality models

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MODELS FOR TIME SERIES ANALYSIS

ZERO MEAN MODELS

- The models have a constant mean and constant variance and shows no predictable trends or seasonality.
- **Independent and identically distributed (iid)** observations
- Represent the random noise around a fixed mean.
- Let $X_1, X_2, \dots, X_n$ represent the random variables corresponding to n observations $x_1, x_2, \dots, x_n$ are n observations from the zero mean time series model.
- The joint distribution of the observations = a product of probability mass function for every time index.

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = f(X_1 = x_1) f(X_2 = x_2) \dots f(X_n = x_n)$$

- $f(X_t = x_t)$ is modeled by a normal distribution of mean zero and variance σ^2
→ assumed to be the irreducible error of the model and hence treated as a random noise.

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MODELS FOR TIME SERIES ANALYSIS

ZERO MEAN MODELS

- The unpredictable portion of the data is treated appropriately.
- Zero mean with constant variance → represents a random noise that can assume infinitely possible real values → suited for representing irregular variations in the time series of a continuous variable.
- Discrete stochastic process in a time series → If $s_1, s_2, \dots, s_m$ represent finite number of possible values $P(X = s_1) = p_1, P(X = s_2) = p_2, \dots, P(X = s_m) = p_m$ such that $p_1 + p_2 + \dots + p_m = 1$.

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MODELS FOR TIME SERIES ANALYSIS

RANDOM WALK

- The model represents a sum of n iids (Independent and identically distributed observations) which has zero mean and constant variance
- A random walk at time index $t \rightarrow S = x_1 + x_2 + \dots x_n$
- A random walk can be easily reduced to zero mean model by taking differences of the observations from two consecutive time indices $\rightarrow$ because $S_t - S_{t-1} = x_t$ is an iid with zero mean and constant variance

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MODELS FOR TIME SERIES ANALYSIS

TREND MODELS

- The model aims to capture the behavior of the time series $\rightarrow$ the long run trend in the time series that can be fitted as linear regression of the time index.
- When the time series does not exhibit any periodic or seasonal fluctuations
$$\mathbf{x_t = \mu(t) + y_t}$$
- Where $\mu(t) \rightarrow$ time-dependent long run trend of the series, $y_t \rightarrow$ noise with zero mean.
- Make the choice of model using EDA – available choices are:
 - Linear model $\rightarrow \mu(t) = w_t + b$
 - non-linear (quadratic) model $\rightarrow \mu(t) = w_1 t + w_2 t^2 + b$
 - Complex non-linear model $\rightarrow \mu(t) = w_0 t^p + b$
- The weights (w) and bias (b) are estimated using regression $\rightarrow$ with t as the independent variable and x_t as the dependent variable.
- $y_t = x_t - \mu(t) \rightarrow$ irreducible noise $\rightarrow$ represent deviations from the trend

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MODELS FOR TIME SERIES ANALYSIS

SEASONALITY MODELS

- Seasonality reveals as periodic and repetitive fluctuations in a time series
→ hence are modeled as sum of weighted sum of sine waves of known periodicity.

- Seasonality model (harmonic regression model) expressed as

$$x_t = S_t + y_t$$

- Where seasonal variation

$$s_t = \sum_{k=1}^L (w_k \cos \alpha t + v_k \sin \alpha t) + b$$

- Residuals = Original data (x_t) - Trend ($\mu(t)$) - Seasonality (S_t).

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MODELS FOR TIME SERIES ANALYSIS

- A summary of the generic approach of a time series analysis:

- Visualize the data at different granularities of the time index to reveal long run trends and seasonal fluctuations.
 - Fit trend line (using regression) to capture long run trends and plot the residuals to check for seasonality or irreducible error.
 - Fit a harmonic regression model (using trigonometric functions like sine and cosine) to capture seasonality.
 - Plot the residuals left by the seasonality model to check for irreducible error.
- The individual trend and seasonality models can be simple or complex depending on the original time series and the application.

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AUTOCORRELATION AND PARTIAL AUTOCORRELATION

- **Stationary (or weakly stationary) time series** → characterized by a constant mean $E(x_t)$
- **Correlation** that depends only on the time lag between two time steps, but independent of the value of the time step → called as autocorrelation. (eg. weekly test marks – depend on previous/earlier test mark)
- **Autocorrelation** (or serial correlation) measures the correlation between values of a time series at different time lags. It indicates how the current value of a variable relates to its past values, helping to identify patterns, trends, or periodicities in the data → Normalized to the range of -1 to 1.
- The lag → the time difference or delay between two observations being compared in a dataset.
- Autocorrelation at lag k is the correlation between a time series X_t and its lagged version X_{t-k} .
- A function of time lag, independent of time index t → used for making inference about future realization of the time series.

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AUTOCORRELATION AND PARTIAL AUTOCORRELATION

- lag k means comparing a value at time t (X_t) with a value at time $t-k$ (X_{t-k}), where k is the number of time steps in the past.
- Consider a time series of daily temperatures (in degrees Celsius):
- Lag $k=1$ →

Comparing each value with the value 1 day earlier.

The pairs are: (X_2, X_1) , (X_3, X_2) , (X_4, X_3) , (X_5, X_4)

Differences (or relationships) between consecutive days are analyzed.

Day	Temperature (X_t)
1	20
2	22
3	21
4	23
5	24

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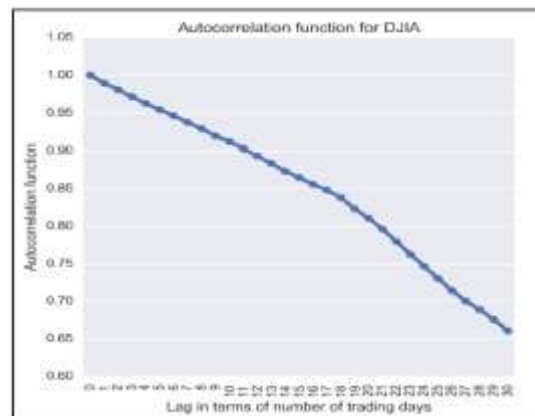
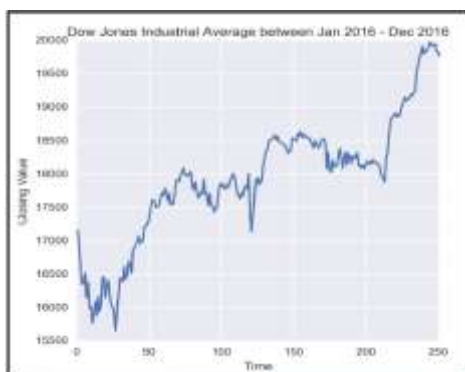
AUTOCORRELATION AND PARTIAL AUTOCORRELATION

- Autocorrelation reflects the degree of linear dependency between the time series at index t and the time series at indices $t-k$ or $t+k$ where k is the lag.
- A positive autocorrelation indicates that the present and future values of the time series move in the same direction
- A negative autocorrelation means that present and future values move in the opposite direction.
- A autocorrelation closer to zero indicates that temporal dependencies within the series may be hard to find.

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AUTOCORRELATION AND PARTIAL AUTOCORRELATION

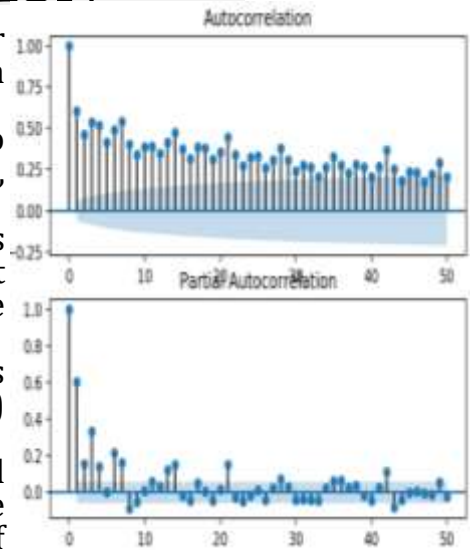
- Presence of autocorrelation identified by plotting the observed values of the **autocorrelation function (ACF)** for a given time series → ACF plot
- **Autocorrelation** → a measure of linear dependency between variables x_t and $x_{(t+h)}$.



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AUTOCORRELATION AND PARTIAL AUTOCORRELATION

- Autoregressive (AR) models capture the linear dependency between variables x_t and x_{t+h} as a linear regression between the x_{t+h} and x_t .
- But in many cases, autocorrelation at lag h is also influenced by the intermediate variables $x_t, x_{t+1}, \dots, x_{t+h-1}$.
- In such cases, partial autocorrelation solves this problem by measuring the correlation between x_t and x_{t+h} when the influence of the intermediate variables has been removed.
- **Partial autocorrelation** in time series analysis defines the correlation between x_t and x_{t+h} which is not accounted for by lags $(t+1)$ to $(t+h-1)$
- **Confidence interval** → a range of values, derived from sample data, that is likely to contain the true population parameter with a specified level of confidence.



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NATURE AND USES OF FORECASTING

- A forecast → a prediction of some future event or events.
- A bad forecast → “The population is constant in size and will remain so right up to the end of mankind” - L’Encyclopedie, 1756.
- In fields including business and industry, government, economics, environmental sciences, medicine, social science, politics, and finance.
- **Forecasting problems** classified as
 1. Short-term → predicting events only a few time periods (days, weeks, and months) into the future.
 2. Medium-term → predicting events from 1 to 2 years into the future
 3. Long-term → predicting events beyond many years
- No dramatic changes in 1st and 2nd types.
- A time series → a time-oriented or chronological sequence of observations on a variable of interest.

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NATURE AND USES OF FORECASTING

- Eg., Many business applications of forecasting utilize daily, weekly, monthly, quarterly, or annual data.
- Instantaneous data, cumulative data, statistic data reflecting an activity during a time period can be observed.
- Forecasting is important in **application areas** as below:
 - ✓ **Operations Management** → to schedule production, control inventories, manage the supply chain, determine staffing requirements, and plan capacity.
 - ✓ **Marketing** → to forecast sales response to advertising expenditures, new promotions, or changes in pricing policies enable businesses to evaluate their effectiveness, determine whether goals are being met, and make adjustments
 - ✓ **Finance and Risk Management** → to forecast returns from their investments and assets that include stocks, bonds, commodities, insurance.
 - ✓ **Economics** → to forecast economic variables, such as gross domestic product, population growth, unemployment, interest rates, inflation, job growth, production, and consumption.
 - ✓ **Industrial Process Control** → to forecast future values of critical quality characteristics of a production process
 - ✓ **Demography** → to forecast births, deaths, and migration patterns of populations stratified by variables such as gender, age, and race.

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NATURE AND USES OF FORECASTING

- 2 types of **forecasting techniques** → Qualitative and Quantitative.
1. **Qualitative forecasting techniques**
 - Subjective in nature and require judgment on the part of experts.
 - Used in situations where there is little or no historical data on which to base the forecast.
 - Eg. the introduction of a new product
 - Basic forecasting methods → the expert opinion of sales and marketing personnel, marketing tests, surveys of potential customers, and experience with the sales performance of other products
 - **Delphi Method** → panel of physically separated experts respond to a **questionnaire** – then a review on predictions relative to the opinions of the entire group → several rounds till convergence to a consensus (or) justified differences of opinion achieved.

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NATURE AND USES OF FORECASTING

2. Quantitative forecasting techniques

- Make formal use of historical data and a forecasting model.
- The model summarizes patterns in the data and expresses a statistical relationship between previous and current values of the variable - Then used to project the patterns in the data into the future - extrapolates past and current behavior into the future.
- Most commonly used models → regression models, smoothing models, and general time series models.
- **Regression models** (or causal forecasting models) → use of relationships between the variable of interest and one or more related predictor variables – **assumed** to have causal relationship – eg. house purchases vs furniture sales
- **Smoothing models** → employ a simple function of previous observations to provide a forecast of the variable of interest - have a formal **statistical basis** – easy to use
- **General time series models** → employ the statistical properties of the historical data to specify a formal model and then estimate the unknown parameters of this model (usually) by least squares.

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NATURE AND USES OF FORECASTING

Forms of the forecast

1. **Point estimate or point forecast** → As a single number that represents the best estimate of the future value of the variable of interest.
 - **Forecast error** → Point estimates are almost always wrong.
 - Hence accompany a forecast with an estimate of how large a forecast error might be → **Prediction Interval (PI)**
2. **PI** → a range of values for the future observation - more useful in decision-making than a single number.
 - **Forecast horizon** (or forecast lead time) → number of future periods for which forecasts must be produced - dictated by the nature of the problem – eg. in production planning, forecasts made on a monthly basis – every month forecast made for the next 3 months → Here Forecast horizon = 3.
 - **Forecast interval** → the frequency with which new forecasts are prepared – eg. 1 month in earlier example.
 - **Rolling or moving horizon forecasting approach** → If the forecast lead time is always the same length, T periods, and the forecast is revised each time period-updates or revises the forecasts for T-1 of the periods in the horizon and computes a forecast for the newest period T.

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FORECASTING PROCESS

The activities in the forecasting process:

1. Problem definition
2. Data collection
3. Data analysis
4. Model selection and fitting
5. Model validation
6. Forecasting model deployment
7. Monitoring forecasting model performance

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FORECASTING PROCESS

1. Problem Definition

- The developing understanding of how the forecast will be used along with the expectations of the customer / user of the product.
- Most important phase of forecasting process in ultimate success of the forecasting model
- Questions that must be addressed during this phase include
 - The desired form of the forecast.
 - The forecast horizon or lead time,
 - How often the forecasts need to be revised (the forecast interval)
 - What level of forecast accuracy is required in order to make good business decisions.
 - Deep understanding on many aspects of the business system.
 - Level of aggregation of the forecast (population level / individual level) must also be considered.

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FORECASTING PROCESS

2. Data Collection

- The process of obtaining the **relevant** history for the variable(s) that are to be forecast, including historical information on potential predictor variables.
- Not all historical data are useful for the current problem → so information collection and storage methods and systems change over time.
- Missing values of some variables, potential outliers, or other data-related problems must be dealt properly.
- Planning done regarding how the data collection and storage issues in the future will be handled so that the reliability and integrity of the data will be preserved.

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FORECASTING PROCESS

3. Data Analysis

- An important preliminary step.
- Purpose → to obtain some “feel” for the data, and a sense of how strong the underlying patterns such as trend and seasonality are.
- Recognizable patterns, such as trends and seasonal or other cyclical components visually inspected from constructed time series plots.
- A trend → evolutionary movement, either upward or downward, in the value of the variable → long-term or more dynamic and of relatively short duration.
- Seasonality → the component of time series behavior that repeats on a regular basis, such as each year.
- Data smoothing performed; Numerical summaries computed; potential outliers should be identified and flagged.
- Helps to suggest the initial types of quantitative forecasting methods and models to explore

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FORECASTING PROCESS

4. Model Selection and Fitting

- Choosing one or more forecasting models and fitting the model to the data.
- Fitting → estimating the unknown model parameters, usually by the method of least squares.
- Several types of time series models available → appropriate model chosen.
- Eg. AR, MA, ARMA, ARIMA, SARIMA, VAR, NN, Spectral Analysis Models, Bayesian Time Series Models
- Assumptions made (hypotheses).
- Evaluation of the quality of the model fit → useful in discriminating between different candidate models.

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FORECASTING PROCESS

5. Model Validation

- **Evaluation** of the forecasting model to determine how it is likely to perform in the intended application
- Examine what magnitude of **forecast errors** will be experienced when the model is used to forecast fresh or new on unseen data.
- Widely used method → data splitting → data are divided into two segments - a **fitting segment** (training data) and a **forecasting segment** (testing data).
- The model is fit to only the fitting data segment, and then forecasts from that model are simulated for the observations in the forecasting segment.
- To ensure that the forecasting model generalizes well and provides accurate predictions for future, unseen data.

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FORECASTING PROCESS

6. Forecasting-Model Deployment

- Getting the model and the resulting forecasts **in use** by the customer.
- Ensure that the **customer** understands **how to use** the model and that generating timely forecasts from the model becomes as **routine** as possible.
- **Model maintenance** → making sure that data sources and other required information will continue to be **available** to the customer.

7. Monitoring Forecasting-Model Performance

- An **ongoing activity** after the model has been deployed to ensure that it is still performing satisfactorily.
- Model **may deteriorate** in performance → will result in larger or more systematic forecast errors.
- Hence monitoring of forecast errors is an essential part of good forecasting system design.
- **Control charts of forecast errors** → track prediction mistakes over time → a simple but effective way to routinely monitor the performance of a forecasting model.

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FORECASTING PROCESS

Control charts of forecast errors → track prediction mistakes over time → a simple but effective way to routinely monitor the performance of a forecasting model.

Used to

- Monitor the consistency of forecasting performance.
- Detect unusual or systematic forecasting errors.
- Identify when the forecasting model needs improvement.

Components of a Control Chart

- X-axis: Time period (Day, Week, Month, etc.)
- Y-axis: Forecast Error (Actual – Forecast)
- Center Line (CL): Mean forecast error (usually close to 0)
- Upper Control Limit (UCL): Maximum acceptable error
- Lower Control Limit (LCL): Minimum acceptable error
- Errors within UCL and LCL: Forecast model is performing normally (in control).
- Errors outside UCL or LCL: Indicates abnormal variation; the forecasting model should be reviewed or updated.
- Continuous upward or downward trend: Suggests bias in the forecasting model.

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DATA FOR FORECASTING

- The Data Warehouse
- Data Cleaning
- Imputation

The Data Warehouse (D/W)

- The output of a forecasting system → could be thought of as **information**, and that output uses **data** as an input.
- D/W → an integrated system comprised of components that are physically and often geographically distributed → a **repository** to store data regarding sales, transactions, company financial and business performance, supplier performance, and customer activity and relations.
- Data loaded in d/w after Data extraction and Data transformation (or data cleaning)

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DATA FOR FORECASTING

The Data Warehouse

Data Extraction

Obtaining data from internal sources and from external sources such as third party vendors or government entities and financial service organizations.

Data Transformation (or data cleaning)

Applying rules to prevent duplication of records and dealing with problems such as missing information

Data Loading

Storing the data into the data warehouse where they are available for modeling and analysis

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DATA FOR FORECASTING

Dimensions of data quality

- Accuracy, timeliness, completeness, representativeness, and consistency

Accuracy

- How close that data conform to its real values.

Timeliness

- Data are as current as possible → periodical updating → the time between the occurrence of the real-world event and its entry into the data warehouse must be as short as possible

Completeness

- The data content is complete, with no missing data and no outliers.

Consistency

- How closely data records agree over time in format, content, meaning, and structure.

Representativeness

- The output data that reside in the data warehouse are similar to the output of a manufacturing process, where the raw data are the input

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DATA FOR FORECASTING

Data Cleaning

- The process of examining data to detect potential errors, missing data, outliers or unusual values, or other inconsistencies and then correcting the errors or problems that are found.
- **Errors** due to the recording or transmission problems → can be corrected by working with the original data source to correct the problem.
- Effective data cleaning – improves the forecasting process.
- Few **automated checks** made:
 - 1. Is there missing data?
 - 2. Does the data fall within an expected range?
 - 3. Are there potential outliers or other unusual values?
- The **rules** employed should be periodically evaluated.
- Use of **graphical displays** assists in identifying unusual data → time series plots, histograms, and scatter diagrams.

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DATA FOR FORECASTING

Imputation

- The process of correcting missing data or replacing outliers with an estimation process → imputation.
- Replaced with a “likely” value based on other available information.

Various Imputation Techniques:

- **Mean value imputation** → replacing a missing value with the sample average calculated from the non-missing observations → easy, but may not be useful.
- **Stochastic mean value imputation** → random variable is added to the mean value to capture some of the noise or variability in the data.

Imputation
Techniques

Mean value
imputation

Stochastic
Imputation

Centered
Moving Average
Imputation

Regression
Imputation

Hot Deck
Imputation

Cold Deck
Imputation

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DATA FOR FORECASTING

Imputation

- **Centered moving average imputation** → A variation of mean value imputation → use a subset of the available historical data that reflects any trend or seasonal patterns in the data.

$$y_j^* = \frac{1}{2k} \left(\sum_{t=j-k}^{j-1} y_t + \sum_{t=j+1}^{j+k} y_t \right)$$

- Eg. k decided based on the seasonal variability in the data, 12 (months)
- y_{11} is found using the sum of y_8 to y_{10} and y_{12} to y_{14} , with $k=3$.
- **Regression imputation** → a variation of mean value imputation → the imputed value is computed from a model used to predict the missing value – linear regression or time series model.
- **Hot deck imputation** → last value carried forward method → missing data is filled using values from a similar record in the same dataset.
- **Cold deck imputation** → missing data is filled using values from an external source or a predefined value (e.g., historical data or domain knowledge).

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RESOURCES FOR FORECASTING

- Many good resources available that can be helpful to technical professionals involved in developing forecasting models and preparing forecasts.
- Three professional **journals** on 'forecast': → publish new methodology, studies devoted to the evaluation of current methods for forecasting, and case studies and applications.
 - Journal of Forecasting
 - International Journal of Forecasting
 - Journal of Business Forecasting Methods and Systems
- Statistics and operations research/management science journals on 'forecast':
 - ✓ Journal of Business and Economic Statistics
 - ✓ Management Science
 - ✓ Naval Research Logistics
 - ✓ Operations Research
 - ✓ International Journal of Production Research
 - ✓ Journal of Applied Statistics

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RESOURCES FOR FORECASTING

- Several **books** based on research on forecasting → they focus on a specific type of forecasting model (AR or ARIMA).
- Statistics **software packages** with very good capability for fitting a variety of forecasting models and solving forecasting problems.
 - ☐ Minitab® Statistical Software
 - ☐ JMP®
 - ☐ The Statistical Analysis System (SAS)
 - ☐ R
 - ☐ Matlab
 - ☐ S-Plus

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THANK YOU



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